



PinchPilot

Small hand movements for webcam desktop control

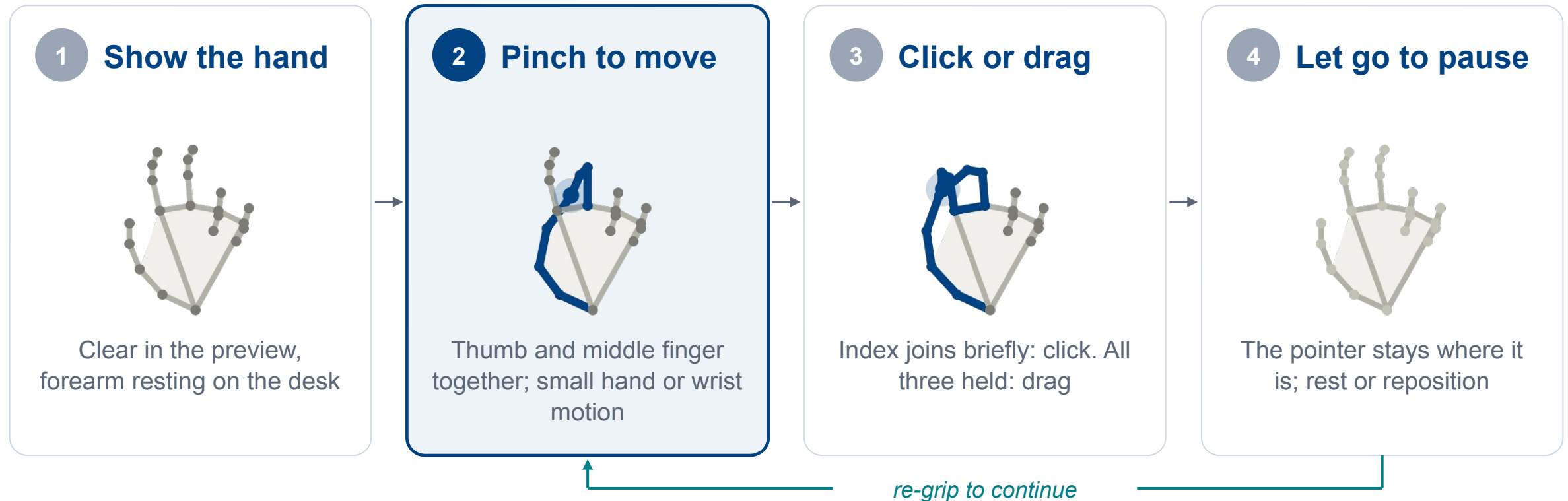
PRS Practice Module · First Project Presentation · Group 15

HE ZIHANG · WEI YI · CHI YONGJUN

6 October 2026

What PinchPilot will do

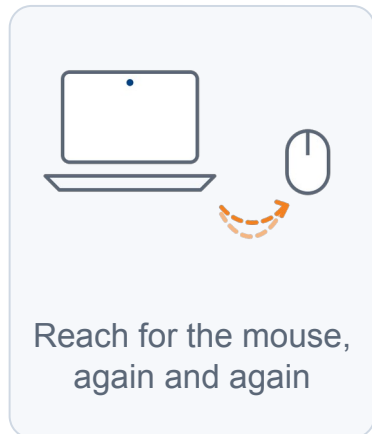
A webcam can see a hand, but that does not make it a comfortable pointer. PinchPilot will let laptop users point, click, drag and scroll with small hand movements and finger pinches.



An ordinary laptop webcam · macOS and Windows · used alongside the mouse or trackpad

Why webcam pointing is hard to use today

A student is reading lecture notes on a laptop. They want to follow a link, scroll on and keep reading, without reaching for the mouse each time.



With a webcam today	What goes wrong	Why
Moving the pointer across the screen	The arm gets tired	Large movements held in the air [1]
Making a small correction	The pointer shakes	Tracking jitter can be as large as the correction
Pinching to click	The click lands off target	The click gesture itself moves the hand

A gesture demo is not yet a daily tool: users must be able to correct, click, rest and resume without the pointer jumping.

[1] Hincapié-Ramos et al., *Consumed Endurance: a metric to quantify arm fatigue of mid-air interactions*, CHI 2014

Project objectives

1 DESIGN

A low-effort interaction

Supported forearm and small movements; the pointer moves only while thumb and middle finger touch.

2 BUILD

A desktop tool

One application for macOS and Windows: preview, sensitivity, gesture switches, practice area and a stop control.

3 LEARN

Reliable contact recognition

Record our own gesture data and compare tuned rules, a random forest and a temporal CNN.

4 EVALUATE

Stability, errors and effort

Recognition on a person the model has never seen, then an internal usability pilot.

OUR MAIN QUESTION

Does learned contact recognition stay reliable when the hand is relaxed or turned, where fixed distance rules tend to fail?

Planned outputs: a working product and reproducible evidence from a small usability pilot.

Related products and our focus

Existing option	What it offers	PinchPilot's focus
Mouse and trackpad	Familiar, direct desktop input	A complementary touch-free option for short tasks; the mouse stays available
Pawvis (macOS)	Webcam control with configurable and trained gestures	Explicit pinch roles, small hand and wrist motion, and support for macOS and Windows
MediaPipe hand tracking	A building block: 21 hand keypoints per frame	A usable interaction, a contact decision and a desktop experience built around the tracking

This is a positioning comparison. How well each option works will be established with matched user tasks; Pawvis is an optional product reference on the same Mac.

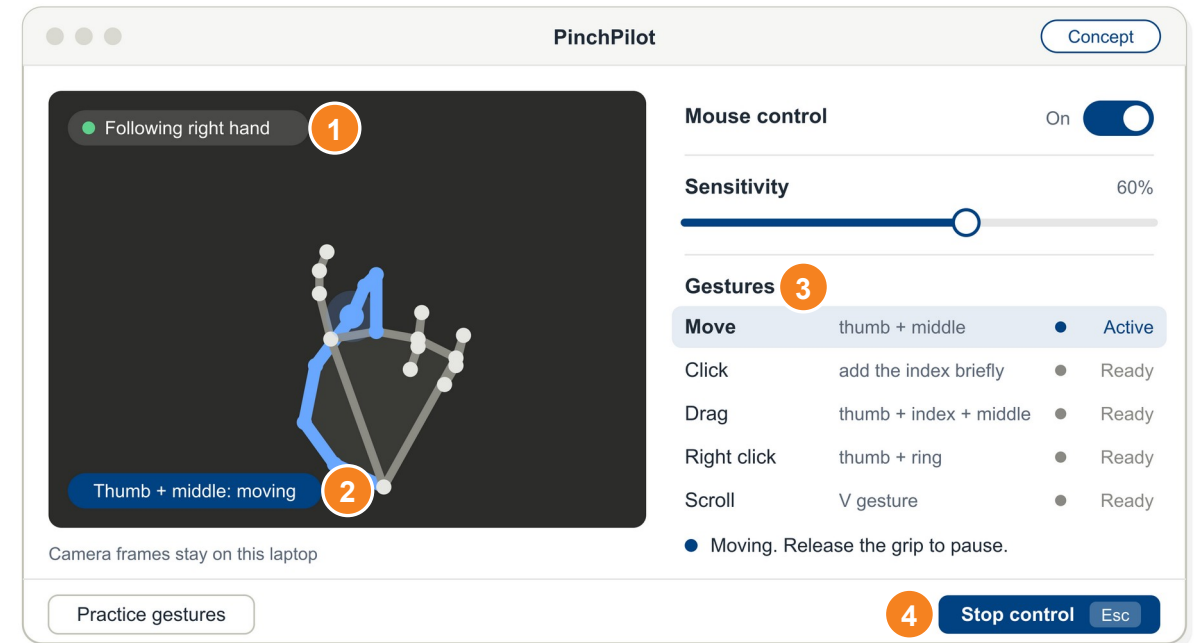
Project scope

IN SCOPE

- One control hand, the primary screen, short seated use
- Point, click, double click, right click, drag, scroll, pause
- An ordinary laptop webcam, on macOS and Windows
- Camera frames are processed on the laptop

OUT OF SCOPE

- Tracking the hand at every angle, or replacing the mouse
- Retraining MediaPipe's hand-tracking model
- Recruited participants: the three of us record the data



- 1 The hand being followed
- 3 Sensitivity and gestures

- 2 The current action
- 4 Stop control and Esc

Concept interface: a design drawing, not a screenshot

Data required and available

What we need

REQUIRED

AVAILABLE NOW

 **Hand keypoints**
From each camera frame

21 points
per frame

Yes
pretrained
MediaPipe

 **Team recordings**
Train and test contact models

3 people
3+ sessions, ~15 min

Not yet
starts 7 Oct

 **Unlabelled usage records**
Pretraining; keypoints only

Keypoints
no video saved

Not yet
after recording starts

 **Synthetic examples**
Software checks only

Checks
never results

Yes
generated

What each recording session will cover

GESTURES

Contact and release

Short taps

Grip plus click

Near-contact (no touch)

Hand adjustments

VARIED ON PURPOSE

Supported posture

Hand orientation

Distance

Lighting

Different days

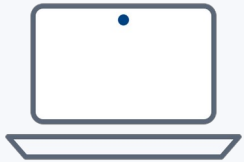
Own laptop camera

Public gesture datasets such as HaGRID label static hand poses, not finger contact over time, so we record our own.

So we will test on a member whose recordings were never used for training, and report the limits of a three-person dataset.

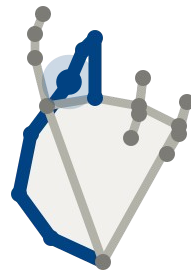
From recordings to training data

ONE MOMENT OF USE



- Laptop camera frames
- About 30 per second
- Processed locally

HAND KEYPOINTS



21 keypoints per frame
from pretrained MediaPipe,
used as it is

FEATURES (SHORT SEQUENCE)

Keypoints relative to the
palm

Three fingertip distances

Time between frames

LABELS (THREE CONTACTS)

- **Thumb and middle**
move
- **Thumb and index**
click
- **Thumb and ring**
right click

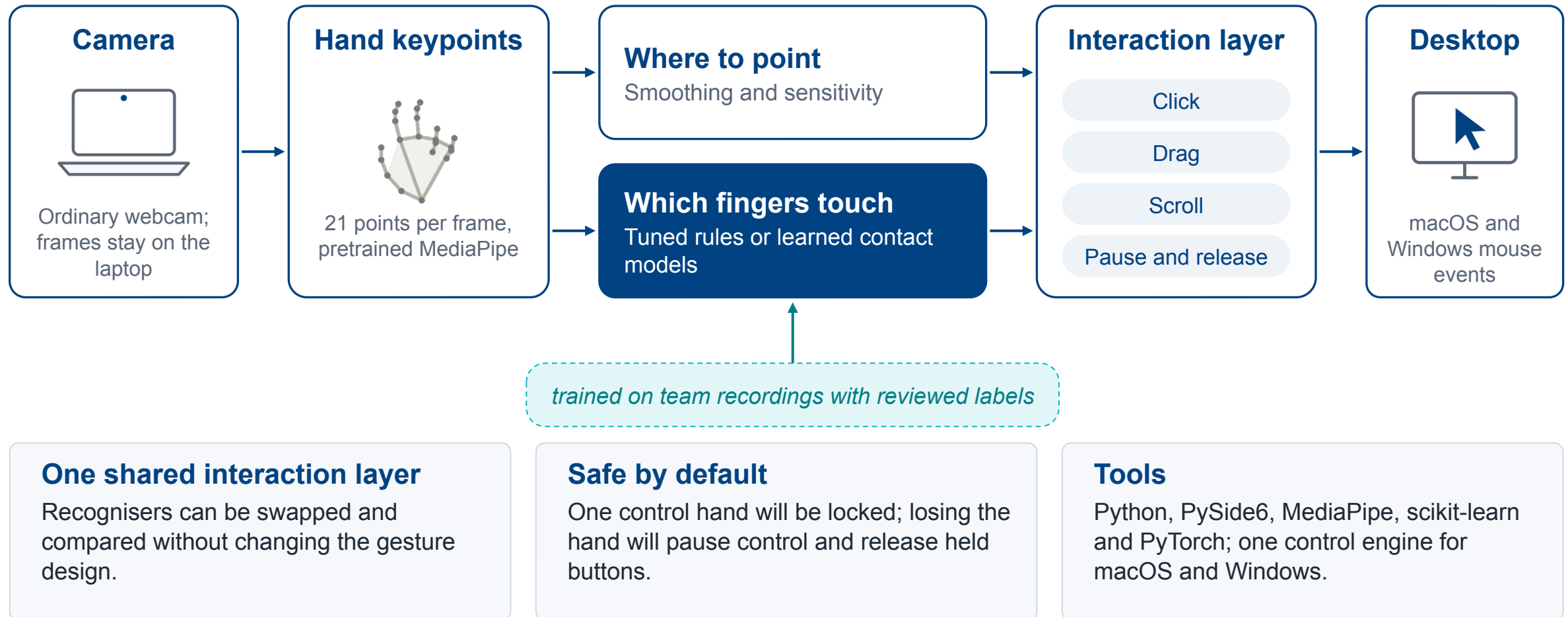
Contacts can be on together: **a drag is the first two, held.**

EXAMPLE The index tip moves to the thumb → the thumb–index distance falls over a few frames → **thumb and index: touching**

Labels prompts organise each trial · the team reviews every label · uncertain moments are marked, not guessed

Labels are never copied from the rules' own output; a second member checks a sample

System design



Pattern recognition techniques

SUPERVISED / UNSUPERVISED

Labels and reconstruction

Reviewed contact labels train the recognisers. An autoencoder learns features from unlabelled keypoint sequences.

MACHINE / DEEP LEARNING

Random forest and temporal CNN

Both read the same short keypoint sequence and output three contact decisions.

HYBRID / ENSEMBLE

Combined methods

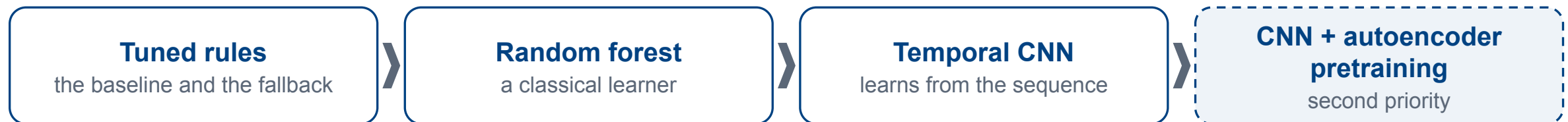
A tree ensemble; pretraining followed by supervised fine-tuning; learned contact with rule-based interaction.

INTELLIGENT SENSING

From camera to action

Camera frames become hand keypoints, hand states, stable movement and desktop actions.

WHAT WE WILL COMPARE, ON THE SAME RECORDINGS



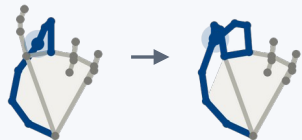
■ **MediaPipe itself is not retrained.** If learning adds little, tuned rules remain a valid fallback.

How one click will be recognised

Worked example

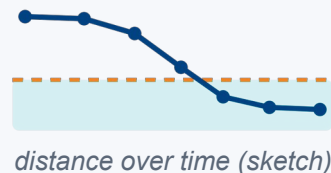
1 Frames

The thumb and middle finger are together; the index finger moves to the thumb.



2 Keypoints

MediaPipe gives 21 keypoints per frame; the thumb–index distance shrinks.



3 Recogniser

Reads the last few frames and outputs three contact decisions.

THUMB WITH

middle · touching

index · touching

ring · apart

4 Interaction

Touch, then release: a click at the held position. Held with the grip: a drag.

short touch → click

all three held → drag

5 Desktop

The operating system receives an ordinary mouse click.



“The camera supplies keypoints, the recogniser decides which fingers touch, and the interaction layer turns that into a deliberate mouse action.”

1 Contact recognition

- Three-fold leave-one-person-out: test on a member never seen in training
- Random forest and CNN against tuned rules, on the same recordings
- Precision, recall, F1; missed and false events; latency

2 Pointer control

- Basic position mapping against stabilised hand and wrist control
- The same recogniser on both sides
- Jitter, click displacement, re-grips and corrections

3 User value

- Internal pilot: three members, three conditions, rotated order
- Models trained without the tester's own recordings
- Time, errors, effort, preference and NASA-TLX (a standard workload questionnaire)

Autoencoder pretraining (second priority): the same CNN with and without it, at 25%, 50% and all labels; results by hand orientation

The test member never guides training or tuning · settings are frozen before each test · three developer-testers give an internal pilot only

Main risks and how we will handle them

1

RISK

Three people are a small dataset

HOW WE WILL HANDLE IT

Test on the member left out of training, report each person separately, and state person and camera effects as limits

2

RISK

Brief contacts are hard to label

HOW WE WILL HANDLE IT

Prompts organise the trials, uncertain moments are marked instead of guessed, and a second member checks a sample

3

RISK

Learning may not beat tuned rules

HOW WE WILL HANDLE IT

Tuned rules remain the fallback; if data collection slips, the autoencoder experiments are deferred first

Limits we will state: three developer-testers; one hand and one screen; MediaPipe's tracking errors are inherited.

What we have done so far



1 Live hand tracking

2 State: grip engaged

3 Gesture switches

4 Practice and recording tools

Screen from our early prototype with the live camera. The interface text is in Chinese for now.

DONE

mostly design

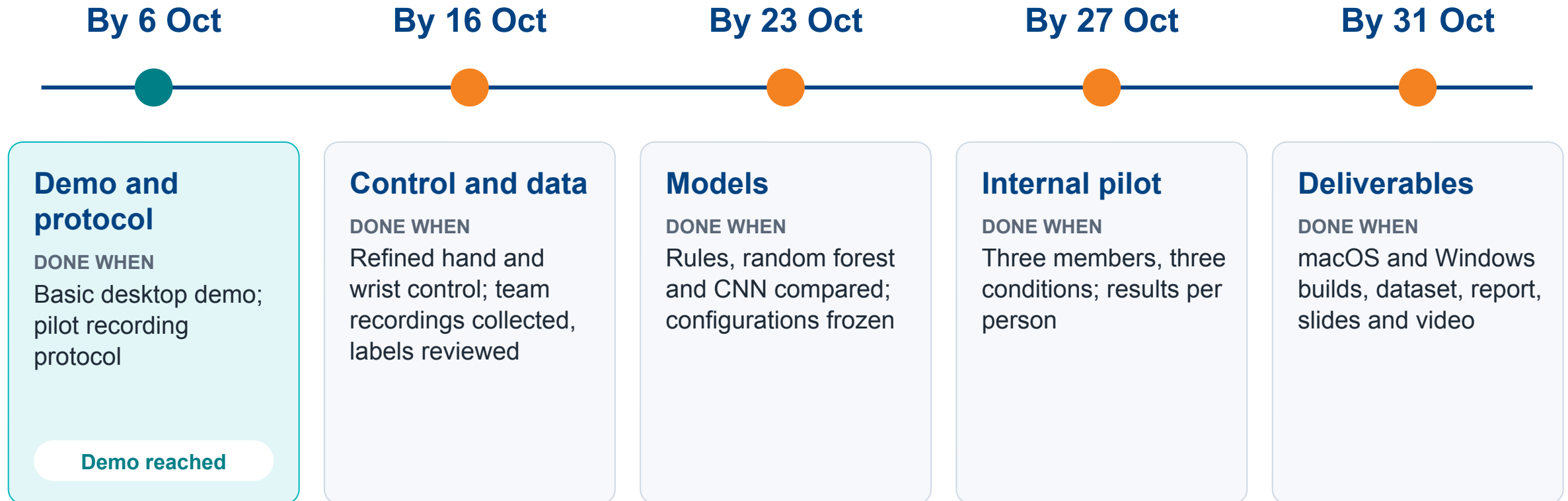
- ✓ **Interaction concept:** supported forearm, small movements, pause and re-grip
- ✓ **Gesture set (provisional):** move, click, right click, drag, scroll, pause; tried in an early prototype
- ✓ **Tools and methods chosen:** MediaPipe keypoints; Python and PySide6; rules, random forest, temporal CNN
- ✓ **Training and evaluation method designed:** leave-one-person-out; checked on synthetic samples only

NOT YET

no accuracy numbers are claimed today

- **Team recordings:** three members, at least three sessions each; recording starts on 7 October
- **Reviewed labels:** the recording protocol is piloted first; a second member checks a sample
- **Model results on real data:** tuned rules, random forest, temporal CNN; autoencoder second
- **Learned control and internal pilot:** a learned recogniser in live use; three members, three conditions

Work plan and next steps



We would welcome your feedback on the data plan and these priorities.

AI tool declaration

OpenAI Codex and Anthropic Claude assisted with source research, proposal drafting, translation, review, figure design and document layout, and with preparing this presentation. The interface on slide 6 is a concept drawing; the screen on slide 14 is captured from our own early prototype, which was also developed with AI coding assistance.

The team is responsible for all content.

Appendix A: evaluation details

Area	Baseline or comparison	Main measures
Contact recognition	Random forest and CNN against tuned geometric rules, on the same held-out recordings	Precision, recall, F1; missed and false events; unavailable input
Autoencoder pretraining	The same CNN with and without pretraining, at three label budgets	F1, missed and false events, latency; results by posture group
Pointer control	Basic mapping against stabilised hand and wrist control; same recogniser	Jitter, click displacement, movement, re-grips and corrections
User tasks	Basic rules; stabilised rules; stabilised control with learned recognition	Completion, time, errors, effort, NASA-TLX and preference
Product reference	Mouse or trackpad; optional Pawvis on the same Mac	Task performance and experience across products

Recognition: three-fold leave-one-person-out; pretraining uses only the training partition; settings are frozen before each test.

Appendix B: proposed gestures

Function	Proposed gesture	Design purpose
Point	Hold thumb and middle finger together, then move the hand or wrist slightly	Small movements with deliberate control
Click and double click	Touch thumb and index finger, then release; repeat for a double click	Active clicks with less target displacement
Right click	Briefly pinch thumb and ring finger	A distinct action for context menus
Drag	Keep the thumb–middle grip and add the index to the thumb; after a short hold, move; release the index to drop	A familiar hold, move and release sequence
Scroll	Form a V with index and middle fingers and move slightly up or down	Read long pages while the pointer stays still
Pause and reposition	Release the thumb–middle grip, reposition and re-grip	Rest, or return inside the camera view